1.28 Graph Transformations: Single Transformations

Worked Solutions

1 a)

Translation by the vector $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$.

1 b)

Translation by the vector $\begin{pmatrix} 0 \\ 8 \end{pmatrix}$.

1 c)

Translation by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$.

1 d)

Translation by the vector $\begin{pmatrix} 0 \\ -11 \end{pmatrix}$.

2 a)

Translation by the vector $\begin{pmatrix} 9 \\ 0 \end{pmatrix}$.

2 b)

Translation by the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$.

2 c)

Translation by the vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

2 d)

Translation by the vector $\begin{pmatrix} 0 \\ -10 \end{pmatrix}$.

2 e)

Translation by the vector $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

2 f)

Translation by the vector $\begin{pmatrix} 0 \\ 8 \end{pmatrix}$.

2 g)

Translation by the vector $\begin{pmatrix} -7 \\ -3 \end{pmatrix}$.

2 h)

Translation by the vector $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$.

3 a)

Translation by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$.

3 b)

Translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

3 c)

Translation by the vector $\begin{pmatrix} 0 \\ -7 \end{pmatrix}$.

3 d)

Translation by the vector $\begin{pmatrix} 0 \\ 15 \end{pmatrix}$.

3 e)

Translation by the vector $\begin{pmatrix} 0 \\ -9 \end{pmatrix}$.

3 f)

Translation by the vector $\begin{pmatrix} 3 \\ 8 \end{pmatrix}$.

4)

$y - 5 = 9(x + 2) + 2 \Rightarrow y - 5 = 9x + 18 + 2$
$\Rightarrow y = 9x + 25.$

5 a)

$y + 7 = 4 - 6(x - 4) \Rightarrow y + 7 = 4 - 6x + 24$
$\Rightarrow y = 21 - 6x.$

5 b)

$y + 7 = (x - 4)^2 \Rightarrow y = (x - 4)^2 - 7.$

5 c)

$y + 7 = (x - 4)^2 + 3 \Rightarrow y = (x - 4)^2 - 4.$

5 d)

$y + 7 = ((x - 4) + 2)((x - 4) - 8)$
$\Rightarrow y = (x - 2)(x - 12) - 7.$

5 e)

$y + 7 = ((x - 4) - 9)((x - 4) + 1)$
$\Rightarrow y = (x - 13)(x - 3) - 7.$

5 f)

$y + 7 = 3^{x-4} \Rightarrow y = 3^{x-4} - 7.$

5 g)

$y + 7 = e^{x-4} - 5 \Rightarrow y = e^{x-4} - 12.$

5 h)

$y + 7 = \frac{1}{x-4} \Rightarrow y = \frac{1}{x-4} - 7.$

5 i)

$y + 7 = \frac{1}{(x-4)^2} \Rightarrow y = \frac{1}{(x-4)^2} - 7.$

5 j)

$y + 7 = \sqrt{x - 4} \Rightarrow y = \sqrt{x - 4} - 7.$

5 k)

$y + 7 = \cos(x - 4) \Rightarrow y = \cos(x - 4) - 7.$

6 a)

Translation by the vector $\begin{pmatrix} 8 \\ 0 \end{pmatrix}$
$\therefore P'(7 + 8, -5) = P'(15, -5).$

6 b)

Translation by the vector $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$
$\therefore P'(7, -5 + 3) = P'(7, -2).$

6 c)

Translation by the vector $\begin{pmatrix} -12 \\ 0 \end{pmatrix}$
$\therefore P'(7 - 12, -5) = P'(-5, -5).$

6 d)

Translation by the vector $\begin{pmatrix} -1 \\ -5 \end{pmatrix}$

$\therefore P'(7-1, -5-5) = P'(6, -10)$.

7)

Translation by the vector $\begin{pmatrix} -2 \\ 2 \end{pmatrix}$

$\therefore y = \mathrm{f}(x+2) + 2$.

8)

Translation by the vector $\begin{pmatrix} -4 \\ -10 \end{pmatrix}$.

9)

Translation by the vector $\begin{pmatrix} 8 \\ -7 \end{pmatrix}$.

10)

Translation by the vector $\begin{pmatrix} -2 \\ -2 \end{pmatrix}$.

11)

$y = \log_5(\frac{x-4}{25}) = \log_5(x-4) - \log_5 25$
$= \log_5(x-4) - 2$

Translation by the vector $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$.

12)

$y = \log_3(9(x-2)) \Rightarrow y = \log_3 9 + \log_3(x-2)$
$\Rightarrow y = 2 + \log_3(x-2)$

Translation by the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$.

13)

$y = \ln(\frac{x+2e}{e^5}) \Rightarrow y = \ln(x+2e) - \ln e^5$
$\Rightarrow y = \ln(x+2e) - 5$

Translation by the vector $\begin{pmatrix} -2e \\ -5 \end{pmatrix}$.

14)

$(x+7)^2 + (y-5)^2 = 9$.

15)

$(x-2)^2 + (y+8)^2 = 3$.

16)

$((x+9)-2)^2+(y+3)^2=10$
$\Rightarrow (x+7)^2+(y+3)^2=10.$

17)

$((x-2)+10)^2+((y+1)-3)^2=16$
$\Rightarrow (x+8)^2+(y-2)^2=16.$

18)

$(x+3)(y-4)+(x+3)^2-(y-4)=10.$

19)

$y-3=(x+2)^3-2(x+2)^2+8(x+2)+2$
$\Rightarrow y=(x+2)^3-2(x+2)^2+8(x+2)+5.$

20)

$y+8=8(x-4)^4-4(x-4)^8$
$\Rightarrow y=8(x-4)^4-4(x-4)^8-8.$

21)

$y+6=\frac{3}{(x+2)^2} \Rightarrow y=\frac{3}{(x+2)^2}-6.$

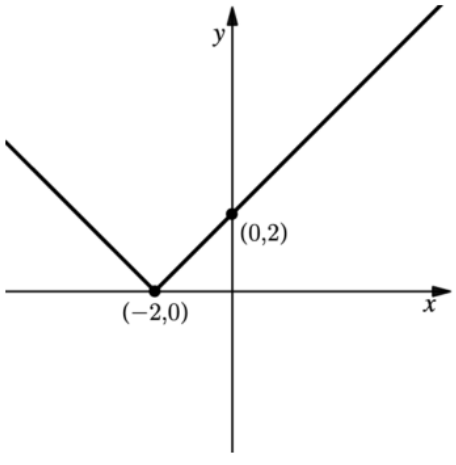
22)

$y=-\frac{4}{(x-7)-7} \Rightarrow y=-\frac{4}{x-14}.$

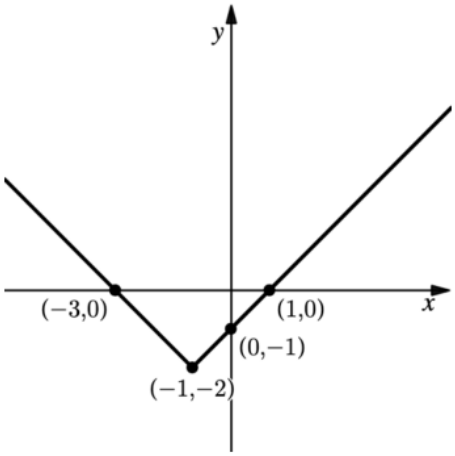
23)

$y=4\times 4^x \Rightarrow y=4^1\times 4^x \Rightarrow y=4^{x+1}$
Translation by the vector $\begin{pmatrix}6\\0\end{pmatrix}$.

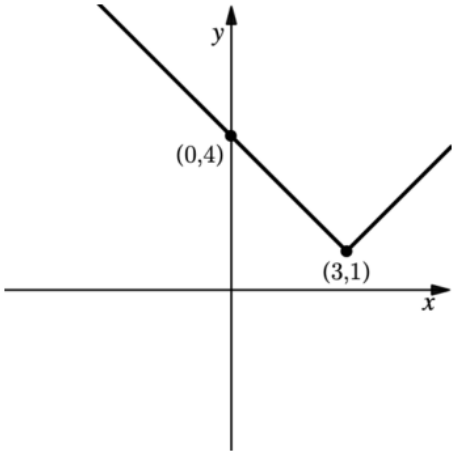
24 a)



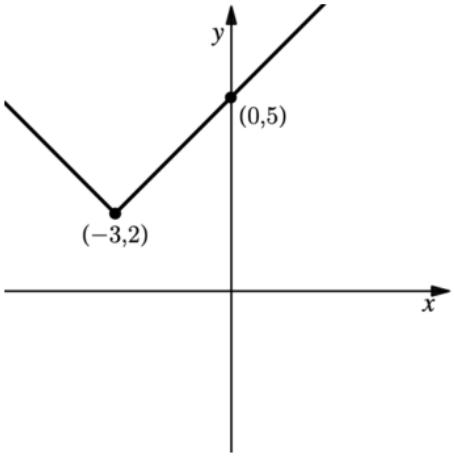
24 b)



24 c)



24 d)



25 a)

Stretch, parallel to the x-axis, scale factor $\frac{1}{5}$.

25 b)

Stretch, parallel to the y-axis, scale factor $\frac{1}{6}$.

25 c)

Stretch, parallel to the x-axis, scale factor $\frac{3}{2}$.

26 a)

$y = 3x^2 - 9 \Rightarrow y = 3(x^2 - 3)$
Stretch, parallel to the y-axis, scale factor 3.

26 b)

$y = 9x^2 - 3 \Rightarrow y = (3x)^2 - 3$
Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$.

27 a)

$y = 8x^3 + 2 \Rightarrow y = (2x)^3 + 2$
Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

27 b)

$y = 8x^3 + 16 \Rightarrow y = 8(x^3 + 2)$
Stretch, parallel to the y-axis, scale factor 8.

27 c)

$y = \frac{1}{8}x^3 + \frac{1}{4} \Rightarrow y = \frac{1}{8}(x^3 + 2)$
Stretch, parallel to the y-axis, scale factor $\frac{1}{8}$.

27 d)

$y = \frac{1}{8}x^3 + 2 \Rightarrow y = (\frac{1}{2}x)^3 + 2$
Stretch, parallel to the x-axis, scale factor 2.

28 a)

$y = 5(4x) + 3 \Rightarrow y = 20x + 3$.

28 b)

$5y = 5x + 3 \Rightarrow y = \frac{1}{5}(5x + 3) \Rightarrow y = x + \frac{3}{5}$.

29 a)

$y = 4 - 6(\frac{1}{3}x) \Rightarrow y = 4 - 2x$.

29 b)

$y = (\frac{1}{3}x)^2 \Rightarrow y = \frac{1}{9}x^2$.

29 c)

$y = (\frac{1}{3}x)^2 + 3 \Rightarrow y = \frac{1}{9}x^2 + 3.$

29 d)

$y = (\frac{1}{3}x + 2)(\frac{1}{3}x - 8)$
$\Rightarrow y = (\frac{1}{3}(x + 6))(\frac{1}{3}(x - 24))$
$\Rightarrow y = \frac{1}{9}(x + 6)(x - 24).$

29 e)

$y = (\frac{1}{3}x - 9)(\frac{1}{3}x + 1)$
$\Rightarrow y = (\frac{1}{3}(x - 27))(\frac{1}{3}(x + 3))$
$\Rightarrow y = \frac{1}{9}(x - 27)(x + 3).$

29 f)

$y = 3^{\frac{1}{3}x}.$

29 g)

$y = e^{\frac{1}{3}x} - 5.$

29 h)

$y = \frac{1}{\frac{1}{3}x} \Rightarrow y = \frac{3}{x}.$

29 i)

$y = \frac{1}{(\frac{1}{3}x)^2} \Rightarrow y = \frac{1}{\frac{1}{9}x^2} \Rightarrow y = \frac{9}{x^2}.$

29 j)

$y = \sqrt{\frac{1}{3}x} \Rightarrow y = \sqrt{\frac{x}{3}} \Rightarrow y = \frac{\sqrt{x}}{\sqrt{3}} \Rightarrow y = \frac{\sqrt{x}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$
$\Rightarrow y = \frac{\sqrt{3x}}{3}.$

29 k)

$y = \cos(\frac{1}{3}x).$

30 a)

$\frac{1}{3}y = 4 - 6x \Rightarrow y = 12 - 18x.$

30 b)

$\frac{1}{3}y = x^2 \Rightarrow y = 3x^2.$

30 c)

$\frac{1}{3}y = x^2 + 3 \Rightarrow y = 3x^2 + 9.$

30 d)

$\frac{1}{3}y = (x + 2)(x - 8) \Rightarrow y = 3(x + 2)(x - 8).$

30 e)

$\frac{1}{3}y = (x-9)(x+1) \Rightarrow y = 3(x-9)(x+1)$.

30 f)

$\frac{1}{3}y = 3^x \Rightarrow y = 3 \times 3^x \Rightarrow y = 3^{x+1}$.

30 g)

$\frac{1}{3}y = e^x - 5 \Rightarrow y = 3e^x - 15$.

30 h)

$\frac{1}{3}y = \frac{1}{x} \Rightarrow y = \frac{3}{x}$.

30 i)

$\frac{1}{3}y = \frac{1}{x^2} \Rightarrow y = \frac{3}{x^2}$.

30 j)

$\frac{1}{3}y = \sqrt{x} \Rightarrow y = 3\sqrt{x}$.

30 k)

$\frac{1}{3}y = \cos x \Rightarrow y = 3\cos x$.

31 a)

$(-2, 6 \times 10) = (-2, 60)$.

31 b)

$(2 \times -2, 10) = (-4, 10)$.

31 c)

$(-2, \frac{2}{5} \times 10) = (-2, 4)$.

31 d)

$(\frac{1}{6} \times -2, 10) = (-\frac{1}{3}, 10)$.

32)

Stretch, parallel to the x-axis, scale factor $\frac{1}{5}$.

33)

Stretch, parallel to the y-axis, scale factor 4.

34)

$y = 2\mathrm{f}(x)$.

35)

Stretch, parallel to the x-axis, scale factor 4.

36 a)

Stretch, parallel to the y-axis, scale factor 27.

36 b)

$y = 27x^3 \Rightarrow y = (3x)^3$
Stretch, parallel to the x-axis, scale factor $\frac{1}{3}$.

37 a)

Stretch, parallel to the y-axis, scale factor 8.

37 b)

$y = \frac{8}{x} \Rightarrow y = \frac{1}{\frac{1}{8}x}$
Stretch, parallel to the x-axis, scale factor 8.

38 a)

Stretch, parallel to the y-axis, scale factor 100.

38 b)

$y = \frac{100}{x^2} \Rightarrow y = \frac{1}{\frac{1}{100}x^2} \Rightarrow y = \frac{1}{(\frac{1}{10}x)^2}$
Stretch, parallel to the x-axis, scale factor 10.

39)

$x^2 + 9y^2 = 16 \Rightarrow x^2 + (3y)^2 = 16$
Stretch, parallel to the y-axis, scale factor $\frac{1}{3}$.

40)

$12x^2 + y^2 = 9 \Rightarrow (2\sqrt{3}x)^2 + y^2 = 9$
Stretch, parallel to the x-axis, scale factor $\frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$.

41)

$y = \mathrm{f}(\frac{1}{2}x)$.

42)

$y = 3((2x) - 2)^2 + 5 \Rightarrow y = 3(2(x - 1))^2 + 5$
$\Rightarrow y = 3 \times 2^2(x - 1)^2 + 5 \Rightarrow y = 12(x - 1)^2 + 5$
Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

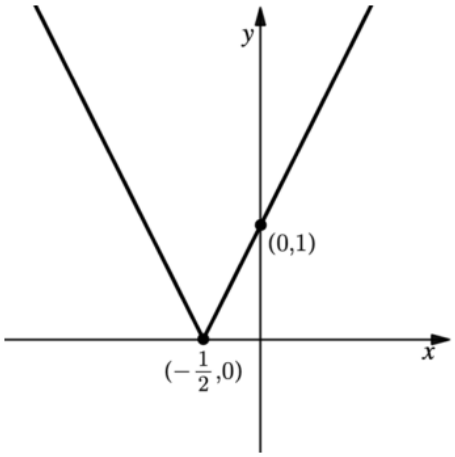
43)

$y = 4((4x) + 8)^2 - 2 \Rightarrow y = 4(4(x + 2))^2 - 2$
$\Rightarrow y = 4 \times 4^2(x + 2)^2 - 2 \Rightarrow y = 64(x + 2)^2 - 2$
Stretch, parallel to the x-axis, scale factor $\frac{1}{4}$.

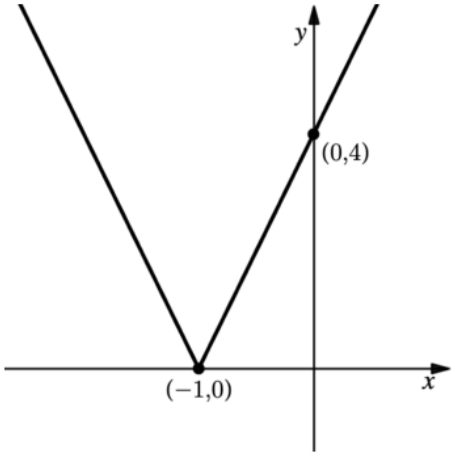
44)

$y = 5((\frac{1}{8}x) - \frac{1}{4})^2 + 1 \Rightarrow y = 5(\frac{1}{8}(x-2))^2 + 1$
$\Rightarrow y = 5 \times (\frac{1}{8})^2(x-2)^2 + 1 \Rightarrow y = \frac{5}{64}(x-2)^2 + 1$
Stretch, parallel to the x-axis, scale factor 8.

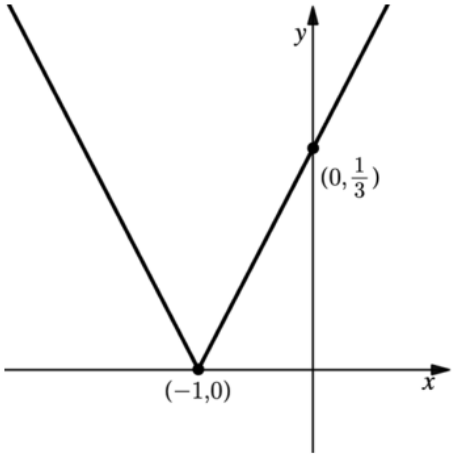
45 a)



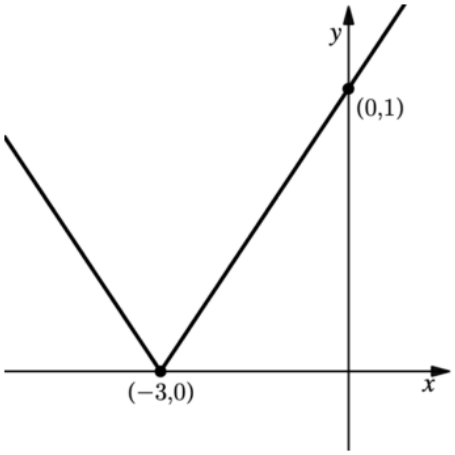
45 b)



45 c)



45 d)



46 a)

Reflection in the y-axis.

46 b)

Reflection in the x-axis.

47 a)

$y = -((x+3)^2 - 4) \Rightarrow y = 4 - (x+3)^2$
Reflection in the x-axis.

47 b)

$y = ((-x)+3)^2 - 4 \Rightarrow y = (3-x)^2 - 4$
Reflection in the y-axis.

48 a)

Reflection in the y-axis.

48 b)

$y = -(e^x - 2) \Rightarrow y = 2 - e^x$
Reflection in the x-axis.

48 c)

$y = \frac{1-2e^x}{e^x} \Rightarrow y = \frac{1}{e^x} - \frac{2e^x}{e^x} \Rightarrow y = e^{-x} - 2$
Reflection in the y-axis.

49 a)

$-y = 8x - 9 \Rightarrow y = -8x + 9 \Rightarrow y = 9 - 8x.$

49 b)

$y = 8(-x) - 9 \Rightarrow y = -8x - 9.$

50 a)

$-y = 3 - 6x \Rightarrow y = -3 + 6x \Rightarrow y = 6x - 3.$

50 b)

$y = 3 - 6(-x) \Rightarrow y = 3 + 6x.$

51 a)

$y = 5 - (-x) \Rightarrow y = 5 + x$
So this function **will not** remain unchanged.

51 b)

$y = (-x)^2 + 2 \Rightarrow y = x^2 + 2$
So this function **will** remain unchanged.

51 c)

$y = (-x)^3 \Rightarrow y = -x^3$
So this function **will not** remain unchanged.

51 d)

$y = 4^{-x}$
So this function **will not** remain unchanged.

51 e)

$y = \frac{1}{(-x)} \Rightarrow y = -\frac{1}{x}$
So this function **will not** remain unchanged.

51 f)

$y = \frac{1}{(-x)^2} \Rightarrow y = \frac{1}{x^2}$
So this function **will** remain unchanged.

51 g)

$y = \sin(-x) \quad (\Rightarrow y = -\sin x)$
So this function **will not** remain unchanged.

51 h)

$y = \cos(-x) \Rightarrow y = \cos x$
So this function **will** remain unchanged.

51 i)

$y = \frac{e^{-x}+e^{-(-x)}}{2} \Rightarrow y = \frac{e^{-x}+e^{x}}{2} \Rightarrow y = \frac{e^{x}+e^{-x}}{2}$
So this function **will** remain unchanged.

52 a)

$-y = 5 - x \Rightarrow y = -5 + x \Rightarrow y = x - 5$
So this function **will not** remain unchanged.

52 b)

$-y = x^2 + 2 \Rightarrow y = -x^2 - 2$
So this function **will not** remain unchanged.

52 c)

$-y = x^3 \Rightarrow y = -x^3$
So this function **will not** remain unchanged.

52 d)

$-y = 4^x \Rightarrow y = -4^x$
So this function **will not** remain unchanged.

52 e)

$-y = \frac{1}{x} \Rightarrow y = -\frac{1}{x}$
So this function **will not** remain unchanged.

52 f)

$-y = \frac{1}{x^2} \Rightarrow y = -\frac{1}{x^2}$
So this function **will not** remain unchanged.

52 g)

$-y = \sin x \Rightarrow y = -\sin x$
So this function **will not** remain unchanged.

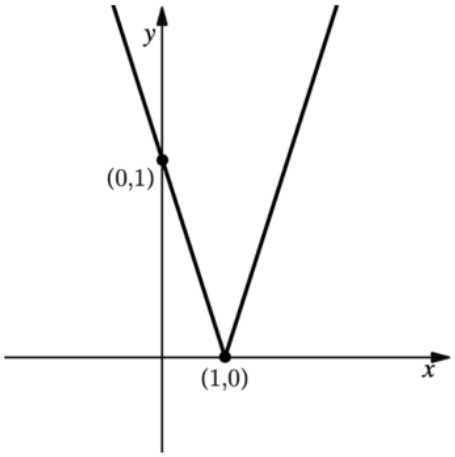
52 h)

$-y = \cos x \Rightarrow y = -\cos x$
So this function **will not** remain unchanged.

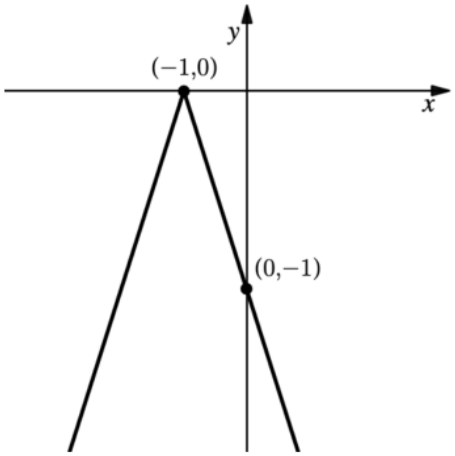
52 i)

$-y = \frac{e^x+e^{-x}}{2} \Rightarrow y = -\frac{e^x+e^{-x}}{2}$
So this function **will not** remain unchanged.

53 a)



53 b)



54 a)

$(4, -(-1)) = (4, 1)$.

54 b)

$(-4, -1)$.

55 a)

Translation by the vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$.

55 b)

$y = \sqrt{(2x)}$
Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$
Alternatively:
$y = \sqrt{2} \times \sqrt{x}$
Stretch, parallel to the y-axis, scale factor $\sqrt{2}$.

55 c)

Translation by the vector $\begin{pmatrix} -9 \\ 0 \end{pmatrix}$.

55 d)

Reflection in the x-axis.

55 e)

$y = 8 \times \sqrt{x}$
Stretch, parallel to the y-axis, scale factor 8
Alternatively:
$y = \sqrt{64} \times \sqrt{x} \Rightarrow y = \sqrt{64x} \Rightarrow y = \sqrt{(64x)}$
Stretch, parallel to the x-axis, scale factor $\frac{1}{64}$.

56)

$\mathrm{f}(x) + 1 = 2x^2 + 1$
$\mathrm{g}(3x) = (3x) - 1 = 3x - 1$
$2x^2 + 1 = 3x - 1 \Rightarrow 2x^2 - 3x + 2 = 0$
$a = 2, b = -3, c = 2$
$\Rightarrow b^2 - 4ac = (-3)^2 - 4 \times 2 \times 2$
$= 9 - 16 = -7 < 0$
$\therefore y = \mathrm{f}(x) + 1$ and $y = \mathrm{g}(3x)$ do not intersect.

57)

$$\begin{aligned} 2f(x) &= 2((x-1)^2 - 1) = 2(x-1)^2 - 2 \\ g(x-2) &= 3(x-2) \\ 2(x-1)^2 - 2 &= 3(x-2) \\ \Rightarrow 2(x^2 - 2x + 1) - 2 &= 3x - 6 \\ \Rightarrow 2x^2 - 4x + 2 - 2 &= 3x - 6 \Rightarrow 2x^2 - 7x + 6 = 0 \\ a = 2,\ b &= -7,\ c = 6 \\ \Rightarrow b^2 - 4ac &= (-7)^2 - 4 \times 2 \times 6 = 49 - 48 = 1 > 0 \end{aligned}$$

$\therefore y = 2f(x)$ and $y = g(x-2)$ intersect at two distinct points.

58)

$y = \frac{9}{16} \times \frac{16}{x+1} = \frac{9}{x+1}$.

Stretch, parallel to the y-axis, scale factor $\frac{9}{16}$.

59 a)

$f(x+2) = \frac{1}{(x+2)^2}$.

Asymptotes at $x = -2$ and $y = 0$.

59 b)

$-f(x) = -\frac{1}{x^2}$.

Asymptotes at $x = 0$ and $y = 0$.

59 c)

$f(x-5) + 4 = \frac{1}{(x-5)^2} + 4$.

Asymptotes at $x = 5$ and $y = 4$.

59 d)

$f(\frac{1}{3}x) = \frac{1}{(\frac{1}{3}x)^2} = \frac{1}{\frac{1}{9}x^2} = \frac{9}{x^2}$.

Asymptotes at $x = 0$ and $y = 0$.

60 a)

$y = f(x-2)$.

Translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

$a = 0,\ b = 7,\ c = 9$.

60 b)

$y = f(x+10)$.

Translation by the vector $\begin{pmatrix} -10 \\ 0 \end{pmatrix}$.

$a = -12,\ b = -5,\ c = -3$.

60 c)

$y = 4f(x)$.

Stretch, parallel to the y-axis, scale factor 4.

$a = -2,\ b = 5,\ c = 7$.

60 d)

$y = f(2x)$.

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

$a = -1,\ b = \frac{5}{2},\ c = \frac{7}{2}$.

60 e)

$y = f(\frac{1}{3}x)$.

Stretch, parallel to the x-axis, scale factor 3.

$a = -6,\ b = 15,\ c = 21$.

60 f)

$y = f(-x)$.

Reflection in the y-axis.

$a = 2,\ b = -5,\ c = -7$.

61 a)

$$\begin{aligned} f(x) &= 0 \\ \Rightarrow (x-2)(x+4)(x+5) &= 0 \\ \Rightarrow x = 2 \text{ or } x = -4 \text{ or } x &= -5 \end{aligned}$$

61 b)

$$\begin{aligned} f(x-2) &= 0 \\ \Rightarrow ((x-2)-2)((x-2)+4)((x-2)+5) &= 0 \\ \Rightarrow (x-4)(x+2)(x+3) &= 0 \\ \Rightarrow x = 4 \text{ or } x = -2 \text{ or } x &= -3 \end{aligned}$$

61 c)

$$\begin{aligned} f(2x) &= 0 \\ \Rightarrow ((2x)-2)((2x)+4)((2x)+5) &= 0 \\ \Rightarrow (2x-2)(2x+4)(2x+5) &= 0 \\ \Rightarrow x = 1 \text{ or } x = -2 \text{ or } x &= -\frac{5}{2} \end{aligned}$$

61 d)

$$\begin{aligned} f(\tfrac{1}{3}x) &= 0 \\ \Rightarrow ((\tfrac{1}{3}x)-2)((\tfrac{1}{3}x)+4)((\tfrac{1}{3}x)+5) &= 0 \\ \Rightarrow (\tfrac{1}{3}x-2)(\tfrac{1}{3}x+4)(\tfrac{1}{3}x+5) &= 0 \\ \Rightarrow x = 6 \text{ or } x = -12 \text{ or } x &= -15 \end{aligned}$$

61 e)

$$\begin{aligned} f(-x) &= 0 \\ \Rightarrow ((-x)-2)((-x)+4)((-x)+5) &= 0 \\ \Rightarrow (-x-2)(-x+4)(-x+5) &= 0 \\ \Rightarrow x = -2 \text{ or } x = 4 \text{ or } x &= 5 \end{aligned}$$

61 f)

$$\begin{aligned} f(x+15) &= 0 \\ \Rightarrow ((x+15)-2)((x+15)+4)((x+15)+5) &= 0 \\ \Rightarrow (x+13)(x+19)(x+20) &= 0 \\ \Rightarrow x = -13 \text{ or } x = -19 \text{ or } x &= -20 \end{aligned}$$

62 a)

$y = \frac{8}{(x+3)-6} \Rightarrow y = \frac{8}{x-3}$.

62 b)

$2y = \frac{8}{x-6} \Rightarrow y = \frac{4}{x-6}$.

62 c)

$y = \frac{8}{(-x)-6} \Rightarrow y = -\frac{8}{x+6}$.

62 d)

$y = \frac{8}{(4x)-6} \Rightarrow y = \frac{4}{2x-3}$.

63)

In order to map $(-3, 0)$ to $(1, 0)$, this would be a translation by the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$. In this case $k = 4$.

In order to map $(6, 0)$ to $(1, 0)$, this would be a translation by the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$. In this case $k = -5$.

In order to map $(8, 0)$ to $(1, 0)$, this would be a translation by the vector $\begin{pmatrix} -7 \\ 0 \end{pmatrix}$. In this case $k = -7$.

64)

In order to map $(-8, 0)$ to $(-1, 0)$, this would be a stretch, parallel to the x-axis, scale factor $\frac{1}{8}$. In this case $k = 8$.

In order to map $(-2, 0)$ to $(-1, 0)$, this would be a stretch, parallel to the x-axis, scale factor $\frac{1}{2}$. In this case $k = 2$.

In order to map $(3, 0)$ to $(-1, 0)$, you could describe this as a stretch, parallel to the x-axis, scale factor $-\frac{1}{3}$. In this case $k = -3$. However, some prefer to describe this as a pair of transformations (in either order), of a stretch, parallel to the x-axis, scale factor $\frac{1}{3}$, and a reflection in the y-axis.

65 a)

Translation by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

65 b)

$y = 4^{x-3} \Rightarrow y = 4^x \times 4^{-3} \Rightarrow y = \frac{1}{64} \times 4^x$.

Stretch, parallel to the y-axis, scale factor $\frac{1}{64}$.

66)

$$\begin{aligned} y = x^2 + 8x - 3 &\Rightarrow y = (x+4)^2 - 16 - 3 \\ &\Rightarrow y = (x+4)^2 - 19 \end{aligned}$$

Translation by the vector $\begin{pmatrix} -4 \\ -19 \end{pmatrix}$.

67)

$$\begin{aligned} y = 2x^2 - 8x - 5 &\Rightarrow y = 2(x^2 - 4x) - 5 \\ &\Rightarrow y = 2((x-2)^2 - 4) - 5 \\ &\Rightarrow y = 2(x-2)^2 - 8 - 5 \\ &\Rightarrow y = 2(x-2)^2 - 13 \end{aligned}$$

Translation by the vector $\begin{pmatrix} 2 \\ -13 \end{pmatrix}$.

68 a)

$$\begin{aligned} y = f(x) &\mapsto y = f(-x) \\ y = 3(-x)^2 - 2(-x) + 1 &\Rightarrow y = 3x^2 + 2x + 1 \end{aligned}$$

68 b)

$$\begin{aligned} y = f(x) &\mapsto y = f(x-3) - 6 \\ y &= 3(x-3)^2 - 2(x-3) + 1 - 6 \\ &\Rightarrow y = 3(x^2 - 6x + 9) - 2x + 6 - 5 \\ &\Rightarrow y = 3x^2 - 18x + 27 - 2x + 1 \\ &\Rightarrow y = 3x^2 - 20x + 28 \end{aligned}$$

68 c)

$$\begin{aligned} y = f(x) \mapsto y &= f(3x) \\ y &= 3(3x)^2 - 2(3x) + 1 \\ \Rightarrow y &= 27x^2 - 6x + 1 \end{aligned}$$

69 a)

$$\begin{aligned} 2x^4 - 7x^3 - 4x^2 &= 0 \\ \Rightarrow (2x^2 - 7x - 4)x^2 &= 0 \\ \Rightarrow (2x+1)(x-4)x^2 &= 0 \\ \Rightarrow x = -\frac{1}{2} \text{ or } x = 4 \text{ or } x &= 0 \end{aligned}$$

69 b)

$$\begin{aligned} x + 3 = -\frac{1}{2} &\Rightarrow x = -\frac{7}{2} \\ \text{or } x + 3 = 4 &\Rightarrow x = 1 \\ \text{or } x + 3 = 0 &\Rightarrow x = -3 \end{aligned}$$

69 c)

$h(x) = 2(3x)^4 - 7(3x)^3 - 4(3x)^2$.

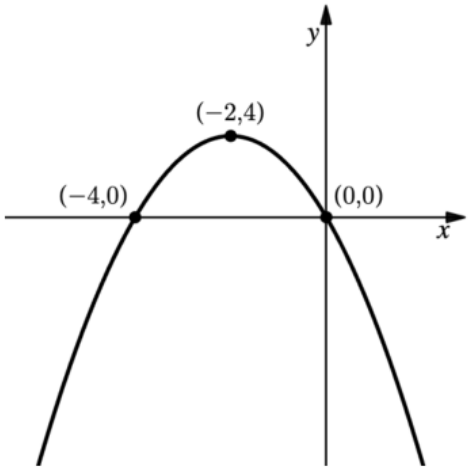
$$\begin{aligned} 3x = -\frac{1}{2} &\Rightarrow x = -\frac{1}{6} \\ \text{or } 3x = 4 &\Rightarrow x = \frac{4}{3} \\ \text{or } 3x = 0 &\Rightarrow x = 0 \end{aligned}$$

69 d)

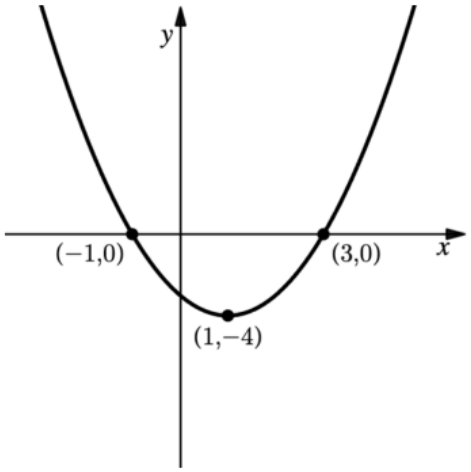
$m(x) = 2(-x)^4 - 7(-x)^3 - 4(-x)^2$.

$$\begin{aligned} -x = -\frac{1}{2} &\Rightarrow x = \frac{1}{2} \\ \text{or } -x = 4 &\Rightarrow x = -4 \\ \text{or } -x = 0 &\Rightarrow x = 0 \end{aligned}$$

70 a)



70 b)



70 c)

